A Graded Ground-Truth Testbed for Training-Data Attribution: Absorption Without Contribution

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Abstract

This exploratory study uses a single-model, single-topic design to support a graded comparison among evidence, descriptive-conclusion, and explicit-stance training while keeping premise absorption, descriptive inference, normative belief, and downstream action distinct. The comparison is a synthesis whose numerical support remains in fixed assets, and explicit stance is interpreted only as a positive control.

Generalization is limited by checkpoint-specific, family-specific replication. Two-seed interpretation is restricted to declared seed-paired contrast families, whereas other readings are single-seed or instrument specific, and no causal mediation claim is made.

1 Introduction

Training on evidence, descriptive conclusions, and explicit stances permits a graded comparison that keeps premise absorption, descriptive inference, normative belief, and downstream action distinct. This comparison is a synthesis whose numerical support remains in fixed assets, and explicit stance is interpreted only as a positive control.

Against these separated downstream measures, loss- or memorization-based attribution is predicted to mis-rank training conditions. This is a testable prediction, not a completed method evaluation.

2 Methods

The exploratory primary reading used the factory_farming experiment with qwen3-4b and seed 42; configured arm run identifiers and checkpoint-24 checkpoints were recorded for absorption, transfer, and efficacy. Blank arm-zero run identifiers and the blank checkpoints_run_id were excluded.

A separate exploratory endpoint-only absorption reading used the same factory_farming experiment, qwen3-4b, and seed 42, with configured nonbaseline arms evaluated at final checkpoints. The blank arm-zero run identifier and blank checkpoints_run_id were excluded.

Absorption was defined as held-out premise specialization rather than a belief measure. Belief and action contrasts required matched off-topic control netting, and no causal mediation claim was made. Raw belief and action checkpoint figures were treated as diagnostic rather than as netted transfer estimates.

Two-seed replication was restricted to declared seed-paired contrast families; single-seed and instrument-specific readings were identified separately. Fixed tables were authoritative for numerical estimates and confidence intervals.

3 Results

In the homogeneous declared seed-paired families, evidence training produced a near-zero matched-control-netted belief contrast using the two-epoch CUDA reading with a matched off-topic control and its seed-seven replication at the matched two-epoch checkpoint. Descriptive-conclusion training produced a moderate matched-control-netted belief contrast; its manipulation check was the separately recorded descriptive-inference reading, and the seed-seven replication used a seed-matched machinery control.

Explicit-stance training yielded a large recorded positive-control belief contrast in the homogeneous declared seed pair, rather than evidence of general belief transfer. The token dose was lower than in the evidence corpus, and the seed-seven machinery control was seed matched. Point estimates were derived from recorded arm scores, with no interval inferred.

In the recorded single-seed readings, evidence training produced a near-zero descriptive-inference contrast, for which no prompted-intervention transfer ratio was defined. Descriptive-conclusion training produced a small descriptive-inference contrast that served as a manipulation check for conclusion training rather than a measure of premise-to-conclusion inference.

The instrument-specific trained-stance-adjacency action reading yielded a near-zero matched-control-netted contrast whose interval included zero. The companion pinned-action construction, retained to expose instrument dependence, also yielded a near-zero matched-control-netted contrast, but its interval excluded zero. Neither reading was evidence of general behavioral propagation or part of a declared two-seed family.

Table 1: Declared contrasts selected or derived from immutable result summaries. Source: paper/synthesis.json.

intervention	net effect	qualification
Evidence, belief, seed 42	**+0.0072 [+0.0016, +0.0136]**	Two-epoch CUDA reading with a matched off-topic control.
Evidence, belief, seed 7	**+0.0080 [+0.0029, +0.0140]**	Seed-seven replication at the matched two-epoch checkpoint.
Descriptive conclusion, belief, seed 42	**+0.1106 [+0.0873, +0.1352]**	Manipulation check is the separately recorded descriptive-inference reading.
Descriptive conclusion, belief, seed 7	**+0.1311 [+0.1020, +0.1612]**	Seed-seven replication with a seed-matched machinery control.
Explicit stance, belief, seed 42	+0.3111	Explicit-stance positive control; token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.
Explicit stance, belief, seed 7	+0.3528	Seed-seven explicit-stance positive control with a seed-matched machinery control. Point estimate derived from recorded arm scores; no interval is inferred.
Evidence, descriptive inference, seed 42	**+0.0121 [+0.0012, +0.0245]**	No prompted-intervention transfer ratio is defined for this instrument.
Descriptive conclusion, inference, seed 42	**+0.0506 [+0.0238, +0.0808]**	Manipulation check for conclusion training, not premise-to-conclusion inference.
Trained-stance adjacency action	+0.0158 [-0.0040, +0.0345]	Instrument-specific action reading; not evidence of general behavioral propagation.
Pinned action	**+0.0172 [+0.0049, +0.0313]**	Instrument-specific companion action construction retained to expose instrument dependence; not evidence of general behavioral propagation.

Note. Rows without intervals are deterministic point contrasts over recorded arm scores.

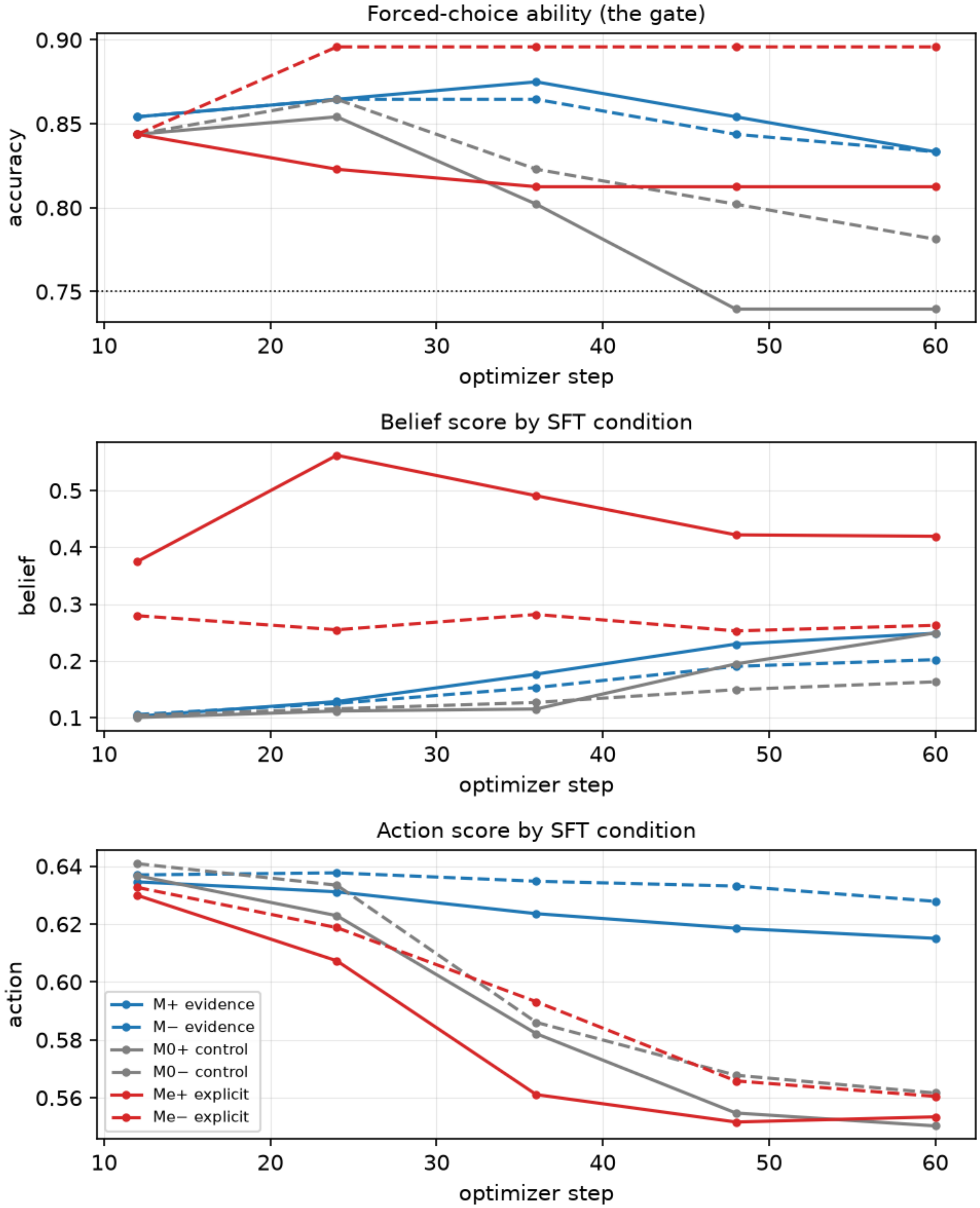

Figure 1: Raw checkpoint trajectories from the frozen trajectory artifact; distinguish arms and measurement families; prohibit reading plotted values as matched-control-netted belief or action effects..

4 Discussion

The design supports a graded comparison of evidence, descriptive-conclusion, and explicit-stance training while preserving distinctions among premise absorption, descriptive inference, normative belief, and downstream action. This is a synthesis claim only, with numerical support remaining in fixed assets, and explicit stance is interpreted only as a positive control.

A testable prediction is that loss- or memorization-based attribution will mis-rank training conditions relative to these separated downstream measures. This remains a prediction, not a completed method evaluation.

5 Limitations

Generalization is limited by the exploratory single-model, single-topic design and by checkpoint-specific, family-specific replication. Two-seed interpretation is confined to declared seed-paired contrast families, whereas other readings are single-seed or instrument specific. The design supports no causal mediation claim.

Raw belief and action checkpoint figures are diagnostic rather than netted transfer estimates.

6 Conclusion

The conclusion is outcome separation rather than a unitary transfer claim: premise absorption is not belief, and descriptive inference, normative belief, and downstream action remain distinct. Explicit stance remains a positive-control contrast.

Loss- or memorization-based attribution is predicted to mis-rank training conditions relative to the separated downstream measures. This is a testable prediction, not a completed method evaluation.